Sepideh Fatemi

+15405585929 | sepidehfaetmi@vt.edu | sepidehfat | sepideh fatemi | sepideh fatemi

Research Interests

Knowledge-Guided Machine Learning, Large Language Models, Symbolic Regression,Time Series Modeling

Education

Virginia Tech — Blacksburg, VA, USA
PhD in Computer Science, GPA: 4.0/4.0 — Jan 2024-present

University of Tehran — Tehran, Iran
B.Sc. in Engineering Science - Computer Specialization, GPA: 3.66/4 — 2017-2022

Research Experience

Modular Symbolic Regression via Vision-Language Guidance — Virginia Tech
Research assistant at Virginia Tech under supervision of Professor Anuj Karpatne — Jan 2025-present

Developing an **VLM & LLM-guided** modular symbolic regression framework to enhance interpretability and scalability, ensuring model-generated equations align with domain knowledge when modelings complex physical systems.

Foundation Model for Domain-Specific Time-Series Data — Virginia Tech
Research assistant at Virginia Tech under supervision of Professor Anuj Karpatne — Jan 2024-present

Developing a Foundation Model for lake ecosystems to generalize across unseen time-series distributions, while also incorporating cross-frequency learning and capturing time-invariant entity characteristics.

Ecological Forecasting using Deep Learning — Virginia Tech
Research assistant at Virginia Tech under supervision of Professor Anuj Karpatne — Jan 2024-present

Predicting chlorophyll-a concentrations using **LSTM**, **Transformer**, and **Transfer Learning** methods.

Modular Compositional Learning by Merging Process-Based Modeling With Deep Learning — Virginia Tech
Research assistant at Virginia Tech under supervision of Professor Anuj Karpatne — Jan 2024-present

Building ML models with Modular Compositional Learning for 2D Lake Hydrodynamics prediction

Using Digital Phenotype to Investigate Cognitive Disorders — University of Tehran
Research assistant at Convergent Technologies Research Center under supervision of Dr. Mohammad Reza Kazemi — Jan 2022-Aug 2022

Developed end-to-end platform for activity classification using Forest and Beiwe libraries.
Developed a back-end and front-end web service displaying diagrams of individuals' activities based on GPS data, which was presented to therapists at the Atieh Clinic.

Behavioral and Neural Analysis of Human Cognitive System — University of Tehran
Internship at Convergent Technologies Research Center under supervision of Dr. Abdol-hossein Vahabie — summer 2021

Conducted research on human cognitive systems, focusing on memory, judgment, and attention bias to design user study for story recall tasks.s
Implemented **speech-to-text**, **Word2Vec**, and **sentiment analysis** tools for NLP tasks involving Persian language text.

Publications

Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance
Sepideh Fatemi, Abhilash Neog, Emma Marchisin, Amartya Dutta , Medha Sawhney , Paul C Hanson, Anuj Karpatne — 2025

CVPPR 2025, CV4Science Workshop (Preprint will be available soon)

Open World Scene Graph Generation using Vision Language Models
Amartya Dutta, Kazi Sajeed Mehrab, Medha Sawhney, Abhilash Neog, Mridul Khurana, **Sepideh Fatemi**, Aanish Pradhan, M Maruf, Ismini Lourentzou, Arka Daw, Anuj Karpatne — 2025

CVPPR 2025, CV in the Wild Workshop (Preprint)

Toward Scientific Foundation Models for Aquatic Ecosystems
Authors: Abhilash Neog, Medha Sawhney, Kazi Sajeed Mehrab, **Sepideh Fatemi**, +10 authors — 2025

ICML 2025, Workshop on Foundation Models for Structured Data
Description: Building a foundation model for lake ecosystems, designed to learn generalizable representations from multi-variable, multi-depth time-series across thousands of lakes.

LakeBeD-US: a benchmark dataset for lake water quality time series and vertical profiles

Authors: Bennett J McAfee, Aanish Pradhan, Abhilash Neog, **Sepideh Fatemi**, Robert T Hensley, Mary E Lofton, Anuj Karpatne, Cayelan C Carey, Paul C Hanson — 2025

Masking the Gaps: An Imputation-Free Approach to Time Series Modeling with Missing Data

Authors: Abhilash Neog, Arka Daw, **Sepideh Fatemi**, Anuj Karpatne — 2024

NeurIPS 2024, Workshop on Time-Series in the Age of Large Models
Description: Incorporating an innovative embedding scheme and a Missing Feature-Aware Attention layer to robustly handle missing data without imputation.

Study of Nonlinear Dynamics and Chaos in Virotherapy and Radio Virotherapy

Authors: **Sepideh Fatemi** - Ehsan Maani Miandoab - Nazanin Imandoost — 2022

The First Conference On Complex Systems With Focus On Network Science
Description: Developed and analyzed mathematical models of virotherapy for cancer treatment, to assess sensitivity and efficiency using numerical methods. Furthermore, we investigated the impact of radiotherapy and key parameters on treatment outcomes.

Teaching Experience

Teaching Assistant – Grad courses: Machine Learning, Artificial Intelligence — Virginia Tech

Teaching Assistant – Undergrad courses: Artificial Intelligence, Advanced Programming, Basic Computer Programming, Database Laboratory — University of Tehran

Selected Projects

AI Art Detection: Distinguishing Human-Created from AI-Generated Artworks — Fall 2024

Built a machine learning model using ViT, ResNet-50, and FFT to classify AI vs. human-made art.

Analyzing the Impact of Client Dropouts on Vertical Federated Learning Models — Fall 2024

Investigated the effect of client dropout patterns on the convergence and accuracy of Vertical Federated Learning (VFL) models.
Implemented neural networks with split learning techniques to analyze feature importance and dropout resilience.
Evaluated optimizer impact (Adam, RMSprop, AdaGrad, SGD) and experimented with feature sampling methods.

Emergence of Social Hierarchies in Multi-Agent Reinforcement Learning — Summer 2023

Developed a multi-agent reinforcement learning environment using OpenAI's Gym and PettingZoo APIs to simulate resource-constrained scenarios.
Utilized Deep Q-Networks (DQN) to train agents, observing the natural development of dominance structures and reduced conflict over time.

Professional Activities

Lighting talk at CVPR2025, CV4Science Workshop — Nashville, TN

Talk on Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance, showcasing a novel framework for modular equation discovery for complex systems.

Participated in Amazon - Virginia Tech Initiative for Efficient and Robust ML Event — September 2024

Engaged in a series of lectures and poster sessions covering state-of-the-art (SOTA) advancements in Machine Learning, with a focus on Large Language Models (LLMs) across various domains.
Interacted with researchers and industry experts to discuss novel ML methodologies and their real-world applications.

Poster Presentation in 6th Iranian Symposium on Brain Mapping Updates — 8 July 2022

Presented my bachelor's thesis project on a smartphone-based digital phenotyping application designed for passive data collection and behavioral analysis using the Beiwe platform.

Member of the Engineering Science Scientific Association, University of Tehran — 2018-2019

Educational and Scientific activities, Certification activities, Competitions and Conferences arrangements

Skills

Programming Languages	Python, Java, C++/C, JavaScript
Machine Learning & Data Science	PyTorch, Scikit-Learn, TensorFlow
Engineering Tools	Matlab, Simulink, Arduino, AutoDesk AutoCad
Web	Spring, Flask, HTML5, CSS, Bootstrap
Database	Postgres, MySQL, MongoDB, Neo4j, ElasticSearch, Kafka